Spatial representations emerge from temporal experience in a hippocampus-inspired navigation agent without spatial supervision

Can spatial representations emerge from sensory inputs and learning objectives that do not explicitly encode space? We investigate this question using a bottom-up computational architecture inspired by hippocampal physiology, without position labels, maps, or spatial rewards. Rather than imposing spatial structure through an explicit learning objective, we explore whether it can emerge from the temporal organization of sensory experience and the agent's interactions with its environment.

Our model combines sparse dentate gyrus (DG)-like sensory responses with CA3-like sequential memory. Temporal relationships between encountered sensory events provide two complementary learning signals: a **"push"** that reinforces DG activity according to the temporal separation between events, and a **"pull"** that motivates the policy to reach internally defined sensory targets. The agent learns from egocentric visual experience and intrinsic rewards, without explicit information about its physical location or spatial distances.

During exploration of a continuous visual environment, spatially informative DG activity emerges. To isolate the contribution of temporal information, we compare temporally weighted DG feedback against mean-matched constant feedback, evaluating the resulting representations on identical sensory observations. Temporal feedback produces higher DG spatial information, less redundant response maps, and a broader spatial distribution of field peaks across five paired training seeds. These improvements extend to downstream decoder representations, where spatial information also increases consistently. In contrast, temporally weighting the policy's target-hit reward does not consistently improve exploration or spatial representation quality compared with constant hit rewards.

The agent's behavior further influences the expression of its learned representations. Even with network weights frozen, the learned policy produces more spatially informative downstream activity than random actions, demonstrating that spatial coding depends not only on sensory representations but also on how an agent interacts with its environment.

Together, these results demonstrate that spatial representations can emerge from temporally structured sensory experience and non-spatial learning objectives in a hippocampus-inspired circuit. They illustrate how temporal learning and self-generated behavior can jointly contribute to spatial organization without explicitly optimizing for space.

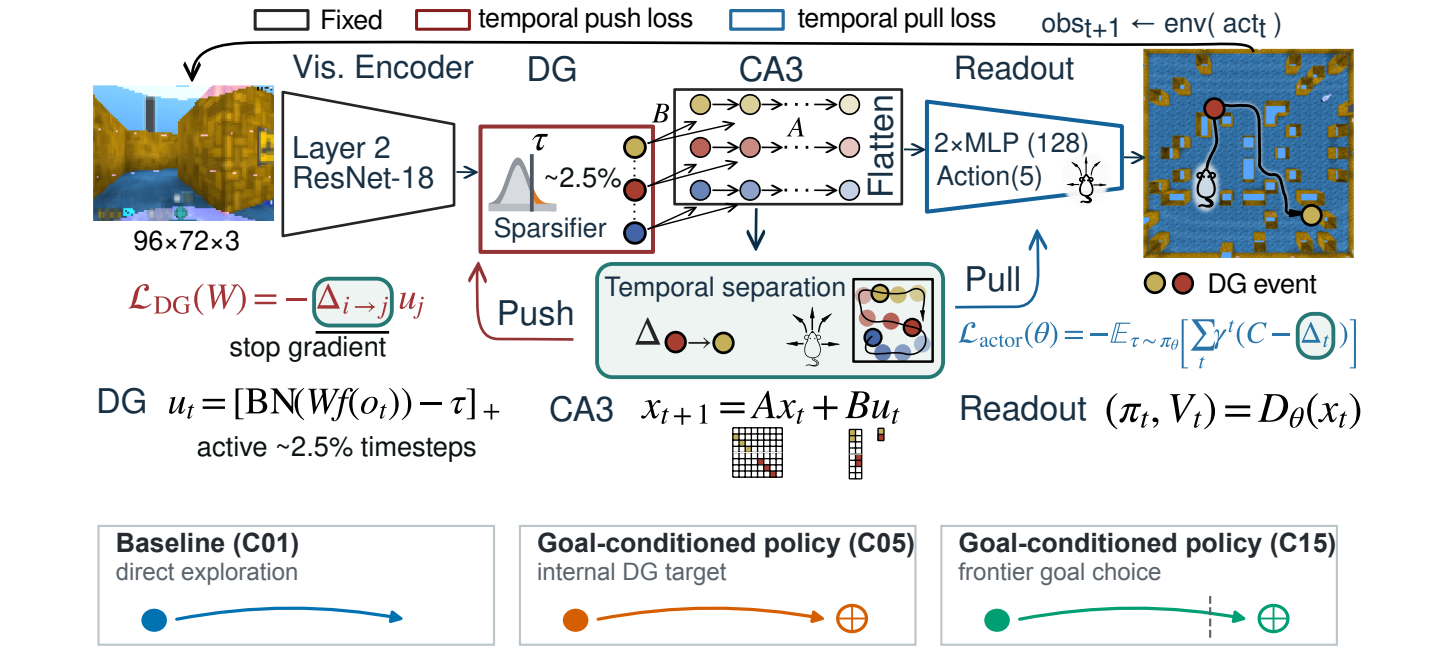


Figure 1: Push–pull architecture and variants. Fixed visual trunk, learned sparse DG, fixed CA3 sequence memory, and policy readout. Red: temporal DG credit; blue: target-hit policy credit. The strip compares baseline, goal-conditioned, and frontier-goal policies.

Spatial emergence and temporal push. DG spatial coding rises during exploration (Fig. 2A). On the same 10,001 observations, temporal DG credit increases the baseline DG spatial score in all five paired seeds (mean +0.0575), reduces map redundancy, and broadens the area covered by eligible field peaks (mean +0.109; Fig. 2C,D). Seed 8 shows the largest peak-area increase among the five pairs. Decoder-1 information also rises by 0.680 bits per activation on average in all five baseline pairs (Fig. 2E). Goal-conditioned comparisons largely point the same way; the frontier condition has only three matched encoder pairs.

Temporal pull. Replacing temporally weighted target-hit credit with constant hit credit has mixed effects on coverage (Fig. 2F) and DG fields. Both rewards still reinforce target arrival, while actor–critic discounting favors earlier arrival. Weakening PPO discounting increases frontier-policy coverage but reduces measured spatial information on separate policy rollouts. The temporal-hit panel remains a placeholder while matched-command and physical-arrival analyses are completed.

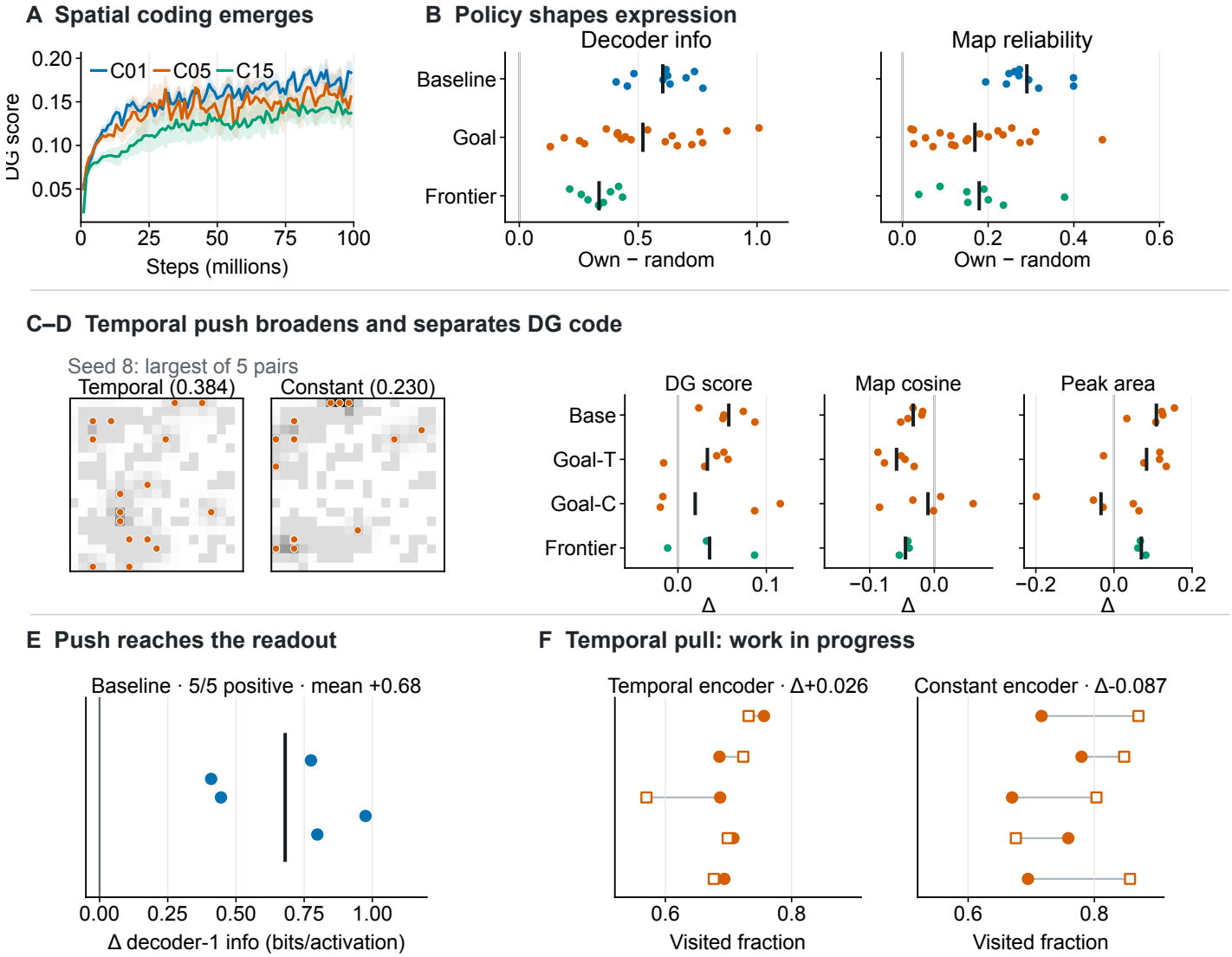


Figure 2: Results grouped by the study's claims. **A,** Online DG spatial score (mean ±1 s.e.m.; five seeds per family). **B,** Decoder-1 information and split-half map reliability under each frozen model's own policy minus random actions (38 models; dots are models, black bars are means). **C,** Baseline seed 8, selected for the *largest* peak-area difference of five paired seeds: temporal 0.384, constant 0.230. Orange dots are eligible peaks after a 0.25% activity cutoff on a shared observation panel; gray bins were unvisited. **D,** All available paired temporal-minus-constant encoder-credit differences in DG spatial score, active-map cosine, and peak area; each dot is a seed, black bars are means. Goal-T and Goal-C fix the temporal or constant worker bonus, respectively. **E,** Baseline decoder-1 information difference on identical observations (5/5 positive). **F,** Work in progress: C05 visited fraction for temporal (filled circles) versus constant (open squares) worker-hit credit, with the encoder held fixed ($n = 5$ per arm); titles give mean paired differences.